

IN VITRO EFFICACY STUDY
EVALUATION OF CELL PROLIFERATION AFTER
TREATMENT WITH COSMETIC PRODUCT

**S & J INTERNATIONAL ENTERPRISES PUBLIC CO.,
LTD.**

KJELDGAARD BARRIER DEFENSE
SBW006-05

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Iscritto al Registro Regione Lombardia ai fini dell'autocontrollo alimentare (N 030015309008)



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REPORT CHANGE RECORD

The table here below reports the change log of all approved changes made to the document that make up the course after initial approval.

Rev. no	Date	Description
00	22/12/2022	Draft report release

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STUDY DESIGN

1.1 Title

In Vitro efficacy study – evaluation of cell proliferation after treatment with a cosmetic product

1.2 Aim the study

Evaluation of cell proliferation after treatment with 3 different concentrations of a product.

The considered cell system is represented by:

- Human skin keratinocytes

1.3 Tested product

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SBW006-05**

Sample description

The product is a cosmetic product with the following INCI composition declared by the Sponsor

Aqua (Water), Glycereth-26, Butylene Glycol, Bis-PEG-18 Methyl Ether Dimethyl Silane, Methyl Gluceth-20, PEG-8, Propanediol, Glycerin, Propylene Glycol, Ethylhexylglycerin, Phenoxyethanol, Bakuchiol, Ferulic Acid, Tocopheryl Acetate, Tromethamine, Carbomer, Bisabolol, Pentylene Glycol, Buddleja Officinalis Flower Extract, Trisodium Ethylenediamine Disuccinate, Squalane, Xanthan Gum, Lecithin, Hydrolyzed Collagen, Hydroxypropyl Cyclodextrin, Alcohol, Polysorbate 20, Retinol, Caprylyl Glycol, Sodium Acetylated Hyaluronate, Sodium Hyaluronate, Sodium Hyaluronate Crosspolymer, Potassium Phosphate, Hydrolyzed Sodium Hyaluronate, Acetyl Hexapeptide-8, BHT, Palmitoyl Tripeptide-38, BHA, Tocopherol, Ascorbyl Palmitate

1.4 Dates

Cell seeding: 28/11/2022

Experimental session: From 30/11/2022 to 02/12/2022

Data analysis: 16/12/2022

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1. MATERIALS AND METHODS

2.1 Monitored experimental conditions

The test was carried out by using cell cultures in the G0 phase of cell cycle in which the sample was added at three different concentrations (0,1%, 0,05% and 0,025% based on a preliminary cytotoxicity test) to study the proliferation activity.

2.2 Test procedure

The proliferation test was carried out by using:

- Human Skin keratinocytes
- Test item concentration: 0,1%, 0,05% and 0,025% in culture medium.
- Human Skin keratinocytes without any treatment as negative control reference (CTR -).

The cells were seeded in a 96 well-plate with incomplete culture medium (missing bovine fetal serum). This induces the phase of cellular quiescence. After 24 hours, the concentrations of the product to be tested were prepared in complete medium, in order to restart the cells from the G0 phase of the cell cycle. The negative control (CTR-) is set up with only complete medium.

Culture medium containing the product at tested concentrations was added to the wells containing cells in the G0 phase of the cell cycle. Cells were exposed to the product for 24, 48 and 72 hours. At the end of incubation period, MTT coloration was performed in order to evaluate cell viability and proliferation index compared to untreated control (CTR-, 100% viability for definition).

Human skin keratinocytes were seeded at three different times (24, 48 and 72 hours) and used as references. Again, MTT coloration was performed to evaluate cell viability and proliferation index compared to untreated control (CTR-, 100% viability for definition).

2.3 MTT assay

MTT-medium is prepared at concentration of 1 mg/ml in culture medium. After exposure of cells to the test items, they were washed with 200 µl of PBS. After removal of the washing solution, 200 µl of MTT-medium are added to each culture well and then incubated for 3 hours at 37°C and 5% CO₂. At the end of the incubation period, the MTT-medium is removed and 200 µl of isopropanol are added.

The plate is shaken on a rotatory plate for 30 minutes, in order to ensure that all the crystals have dissolved from the cells and have formed a homogeneous solution. The absorbance is measured at 570 nm on a microplate reader. The results are expressed as % cell viability compared to an untreated control cell culture.

2.4 References

DB-ALM Method Summary n. 17: MTT Assay – Summary

2.5 Statistical analysis

Statistical analysis by T-test was conducted between the samples and the untreated negative control (CTR -). The variations are considered significant for p<0,05.

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2. RESULTS

In the tables and graphs below the data collected during the test were reported.

Table 1: Cell proliferation study on human skin keratinocyte cell culture (24h, 48h and 72h – KJELDGAARD BARRIER DEFENSE SBW006-05) compared to CTR- (100%).
Significant data vs CTR- are reported with asterisk (*).

KJELDGAARD BARRIER DEFENSE	Cell Viability		Cell Proliferation	
Experimental Time 24h	mean	st.dev.	(CTR- as 100%)	
0,100%	112,1%	0,3%	12,1%	*
0,050%	130,4%	10,7%	30,4%	*
0,025%	141,0%	12,9%	41,0%	*

KJELDGAARD BARRIER DEFENSE	Cell Viability		Cell Proliferation	
Experimental Time 48h	mean	st.dev.	(CTR- as 100%)	
0,100%	109,0%	1,9%	9,0%	
0,050%	120,4%	0,7%	20,4%	*
0,025%	153,9%	4,4%	53,9%	*

KJELDGAARD BARRIER DEFENSE	Cell Viability		Cell Proliferation	
Experimental Time 72h	mean	st.dev.	(CTR- as 100%)	
0,100%	108,4%	2,3%	8,4%	
0,050%	110,2%	0,7%	10,2%	
0,025%	111,6%	11,2%	11,6%	

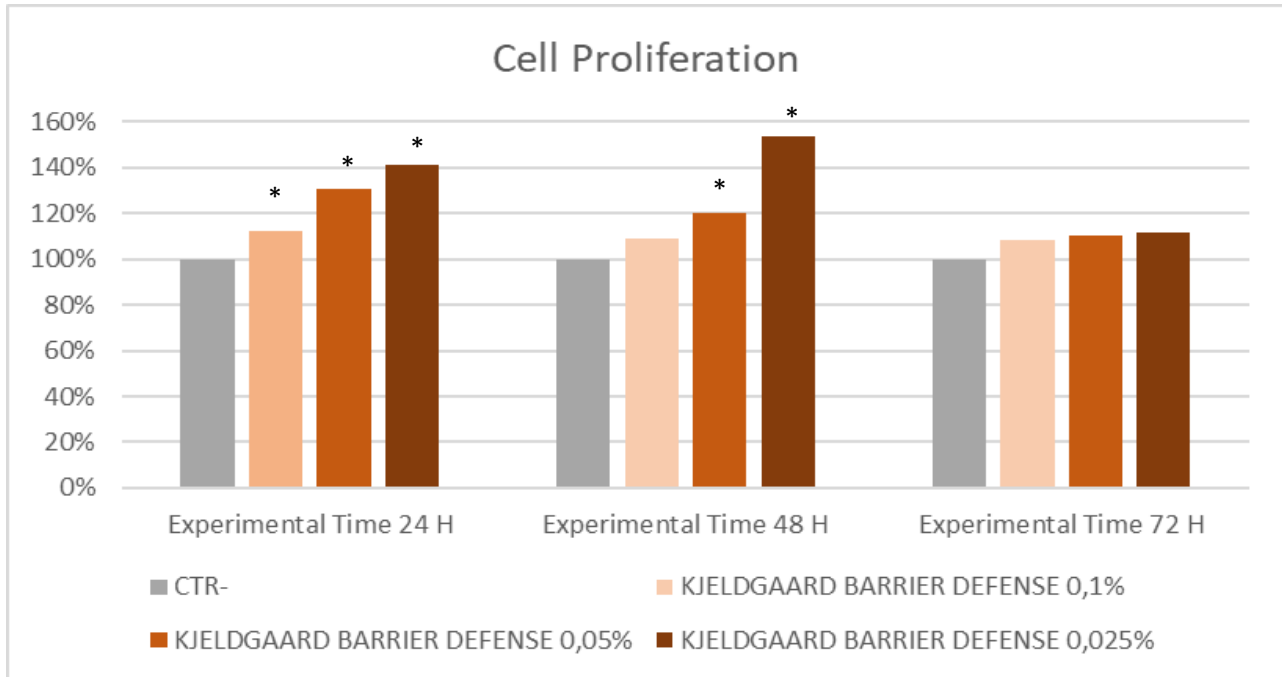
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Graph 1: Cell proliferation study on cell culture (24h, 48h and 72h – KJELDGAARD BARRIER DEFENSE SBW006-05) compared to CTR- (100%).



Comment to the results

The tested product resulted effective in increasing cell proliferation on the considered *in vitro* system. In particular, the tested product at 0,1%, 0,05% and 0,025% enhances significantly cell proliferation after 24 hours. Similarly, at 48 hours cell proliferation is significantly increased at concentrations 0,05% and 0,025%. After 72 hours, no significant increases are shown.

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4. CONCLUSIONS

According to the data obtained and reported in this report, we can assess that the treatment of cell cultures with tested product enhances cells proliferation in the considered *in vitro* system.

According to tested experimental patterns,

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**significantly enhances proliferation capability of cell culture in the specific *in vitro* model,
highlighting a potential in the acceleration of skin renewal on a cellular level**

Experimenter

Dr. Andrea POGGI

Study Director

Dr. Silvana GIARDINA

- ▶ The result of the study reported in this document is only referred to the tested sample and the specific experimental conditions.
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